

SaaS Trial-to-Paid Activation & Churn Analytics

Problem Statement

A SaaS company is experiencing significant drop-offs between free trial signup and paid conversion, leading to lost revenue opportunities. The objective is to identify early behavioral signals and user segments that indicate low conversion likelihood, and recommend targeted interventions to improve trial-to-paid conversion rates.

Assumptions

- The free trial is assumed to be exactly 14 days long.
 - If a user pays for a subscription right after their 14-day free trial ends, they are success(1). If they don't buy the subscription and stop using the app, they are drop-off(0).
 - To find "early warning signals", we will look at what user did during their first 7 days on the platform.
 - Any row in feature_usage.csv where error_count > 0 is assumed to represent a user-facing glitch or friction point that negatively impacts user experience.
 - A first_response_time_minutes of greater than 120 minutes (2 hours) or a satisfaction_score of 1 or 2 stars during the trial is assumed to be a primary driver of conversion failure.
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Hypothesis

- Users are getting so much done during the free trial that their temporary business needs are fully satisfied. Once the trial ends, they don't buy because they don't need the tool anymore.
 - Users lose interest quickly after trying the app in Week 1 and stop logging in during Week 2.
 - If a trial user hits too many technical glitches or user-experience errors in their first week, they will lose trust and abandon the app before converting.
 - If a free-trial user reaches out for help and gets stuck waiting for a response, their momentum dies, and they will abandon the trial.
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Dataset Summary

(Source: Kaggle datasets; SaaS Subscription & Churn Analytics Dataset by RIVER)

The analysis is built on RavenStack, a simulated multi-table relational dataset representing a real-world AI-powered SaaS collaboration platform. Synthetic telemetry generated via Python and ChatGPT provides a realistic environment to execute complex SQL joins, cohort analysis, churn modeling, and revenue tracking while accounting for real-world edge cases.

The dataset spans 5 core relational tables:

File Name	Description & Key Attributes	Primary Analytical Use
accounts.csv	Customer metadata, signup dates, industry verticals, and acquisition channels.	Cohort grouping, acquisition channel performance.
subscriptions.csv	Subscription lifecycles, plan tiers (Basic, Pro, Enterprise), trial flags, and revenue (MRR).	Conversion tracking, tier adoption, lifetime value.
feature_usage.csv	Daily product interaction logs, click counts, session durations, and error events.	Engagement thresholds, feature adoption, bug proxy analysis.
support_tickets.csv	Ticket submission logs, first response times, resolution durations, and CSAT scores.	Support friction evaluation, SLA impact on churn.
churn_events.csv	Churn dates, primary cancellation reasons, and refund request behaviors.	Churn modeling, cancellation driver identification.

Tools Used

- Data Cleaning:** Python (Pandas, NumPy), SQLAlchemy.
- Database Management:** MySQL (Relational Schema, Window Functions, Joins, CTEs)
- Exploratory Data Analysis:** Seaborn, Matplotlib
- Statistical Testing:** scipy.stats
- Business Intelligence:** Power BI (for dashboarding)

Executive Summary

This report presents an end-to-end analytics investigation into the product engagement dynamics, onboarding friction points, support operations, and pricing plan conversions for a B2B SaaS platform. Analyzing 500 enterprise trial accounts over a strict 14-day trial window, the project diagnoses why 68.40% of trial accounts churn and identifies high-leverage user behaviors that drive the baseline 31.60% conversion rate.

Phase 1: Data Wrangling & Feature Engineering

The goal of this phase was to turn raw, messy CSV files into a high-quality "Master Dataset".

1. Data Ingestion

- I developed a Python script using the SQLAlchemy and logging libraries. Instead of a simple copy-paste, I implemented logging to track the success or failure of every row inserted into MySQL. This ensures the pipeline is "Production-Ready."
- To ensure seamless compatibility across the full data stack (**MySQL, Python/Pandas, and Power BI**), I implemented a 'Standardized Naming Convention' using **snake_case**.
- Raw data was first ingested into **SQL** for structured storage. The processed tables were then exported to **Python (Pandas)** for advanced feature engineering.

2. Data Filtering

- To optimize model performance and visual clarity, **non-essential attributes were excluded** during the data transformation phase.
- **Column Retention:** Kept only analytical attributes:
 - **Accounts:** `account_id, industry, signup_date, referral_source, plan_tier, seats, is_trial`
 - **Subscriptions:** `subscription_id, account_id, start_date, plan_tier, mrr_amount, is_trial, billing_frequency`
 - **Feature Usage:** `subscription_id, usage_date, feature_name, usage_count, usage_duration_secs, error_count, is_beta_feature`
 - **Support Tickets:** `account_id, submitted_at, resolution_time_hours, first_response_time_minutes, satisfaction_score`

3. Data Cleaning

➤ *Duplicates Handling*

- Verified that all tables contain 0 duplicate rows.

➤ **Missing Data Imputation**

- When looking at the customer support table, it is found that out of 2,000 submitted support tickets, only 1,175 had a satisfaction score. The remaining 825 rows were completely blank (NaN).
- A user who completely ignores a satisfaction survey after talking to support is often displaying extreme frustration.
- Therefore, I purposely kept those values as NaN to test a new behavioral hypothesis: ***Are trial users who ignore support surveys more likely to abandon the product entirely compared to those who take the time to leave feedback?***

➤ **Data Inconsistencies Audit**

- Date fields across all raw tables (df_accounts, df_subscriptions, df_feature_usage, df_support_tickets) were initially ingested as generic object (string) data types, preventing time-series math.
- Converted all date columns into explicit Pandas datetime64[ns] format (pd.to_datetime).
- Identified a major system tracking issue where **13,198 usage logs and 1,077 support tickets** pre-dated user account registration (signup_date).
- Removed all the product usage logs and support tickets that had dates before the user's official signup date. This ensures our analysis only focuses on real activity that happened while the customer was actually using the free trial.

4. Feature Engineering

A centralized modeling table, master_analytics, was generated by consolidating raw relational sources (df_accounts, df_subscriptions, df_feature_usage, df_support_tickets). To capture engagement, performance, and friction metrics per account within the 14-day trial window, the following aggregate features were computed and integrated into master_analytics:

- **converted_to_paid (Target Variable):** Binary indicator (1 if converted within $0 \leq \text{days} \leq 14$, 0 otherwise).
- **active_days_count:** How many separate days out of the first 7 they actually opened the tool and used it.
- **distinct_features_tried:** Out of the 40 tools available, this counts how many different features they experimented with.
- **total_clicks & total_duration_mins:** Summed engagement intensity metrics.
- **total_errors_encountered:** The total number of system errors or software bugs the user hit in their first 7 days.
- **tickets_raised_count:** The number of times they asked customer support for help during their 14-day trial.

- **avg_response_time_mins:** How many minutes it took our team to reply to their questions.
- **avg_csat_score:** The star rating (1–5) they gave our support team after their issue was resolved.
- **skipped_surveys_count:** The number of times they opened a support ticket but completely ignored the feedback survey at the end.

```
master_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 500 entries, 0 to 499
Data columns (total 16 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   account_id                           500 non-null    object
1   industry                             500 non-null    object
2   signup_date                          500 non-null    datetime64[ns]
3   referral_source                      500 non-null    object
4   plan_tier                           500 non-null    object
5   seats                               500 non-null    int64
6   converted_to_paid                    500 non-null    int64
7   total_clicks                         500 non-null    float64
8   total_duration_mins                 500 non-null    float64
9   total_errors_encountered             500 non-null    float64
10  distinct_features_tried              500 non-null    float64
11  active_days_count                    500 non-null    float64
12  tickets_raised_count                 500 non-null    float64
13  avg_response_time_mins                500 non-null    float64
14  avg_csat_score                       500 non-null    float64
15  skipped_surveys_count                500 non-null    float64
dtypes: datetime64[ns](1), float64(9), int64(2), object(4)
memory usage: 62.6+ KB
```

Re-loaded cleaned datasets into MySQL with strict schema enforcement, explicit data types (INTEGER, VARCHAR, DATETIME, DECIMAL), primary keys, and foreign key relationships.

Phase 2: Exploratory Data Analysis & Hypothesis Testing

Exploratory Data Analysis was conducted prior to modeling and dashboard construction to transform raw event logs and account metrics into validated business signals. Rather than jumping straight to visualizations.

1. Descriptive Statistics, *Outliers*, and *Correlations*

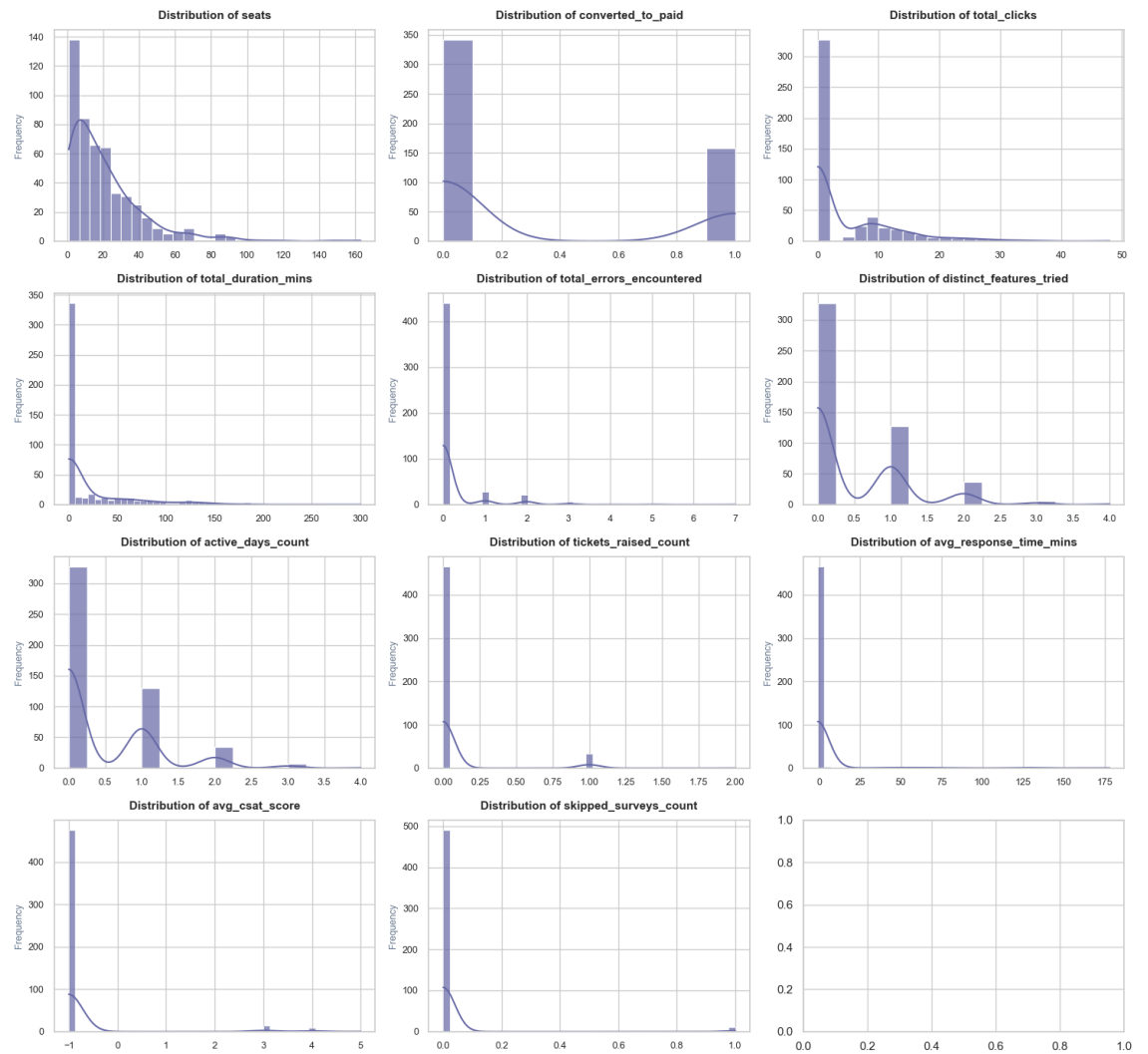
a) Summary Statistics

Used `master_analytics.describe(include='all')` for summary statistics.

master_analytics.describe(include='all')										
Python										
	account_id	industry	signup_date	referral_source	plan_tier	seats	converted_to_paid	total_clicks	total_duration_mins	total_errors_encountered
count	500	500	500	500	500	500.000000	500.000000	500.000000	500.000000	500.000000
unique	500	5	NaN	5	3	NaN	NaN	NaN	NaN	NaN
top	A-00bed1	DevTools	NaN	organic	Pro	NaN	NaN	NaN	NaN	NaN
freq	1	113	NaN	114	178	NaN	NaN	NaN	NaN	NaN
mean	NaN	NaN	2024-01-27 06:34:33.600000256	NaN	NaN	20.560000	0.316000	4.526000	22.030728	0.228000
min	NaN	NaN	2023-01-02 00:00:00	NaN	NaN	1.000000	0.000000	0.000000	0.000000	0.000000
25%	NaN	NaN	2023-08-06 18:00:00	NaN	NaN	5.000000	0.000000	0.000000	0.000000	0.000000
50%	NaN	NaN	2024-02-21 00:00:00	NaN	NaN	15.000000	0.000000	0.000000	0.000000	0.000000
75%	NaN	NaN	2024-08-04 06:00:00	NaN	NaN	28.000000	1.000000	8.250000	24.825000	0.000000
max	NaN	NaN	2024-12-31 00:00:00	NaN	NaN	163.000000	1.000000	48.000000	300.100000	7.000000
std	NaN	NaN	NaN	NaN	NaN	21.044718	0.465379	7.506932	42.980899	0.732864
distinct_features_tried	active_days_count	tickets_raised_count	avg_response_time_mins	avg_csat_score	skipped_surveys_count					
500.000000	500.000000	500.000000	500.000000	500.000000	500.000000					
NaN	NaN	NaN	NaN	NaN	NaN					
NaN	NaN	NaN	NaN	NaN	NaN					
NaN	NaN	NaN	NaN	NaN	NaN					
0.456000	0.450000	0.072000	4.369000	-0.774000	0.02000					
0.000000	0.000000	0.000000	-1.000000	-1.000000	0.00000					
0.000000	0.000000	0.000000	-1.000000	-1.000000	0.00000					
0.000000	0.000000	0.000000	-1.000000	-1.000000	0.00000					
1.000000	1.000000	0.000000	-1.000000	-1.000000	0.00000					
4.000000	4.000000	2.000000	178.000000	5.000000	1.00000					
0.719095	0.704623	0.266379	23.968423	0.996449	0.14014					

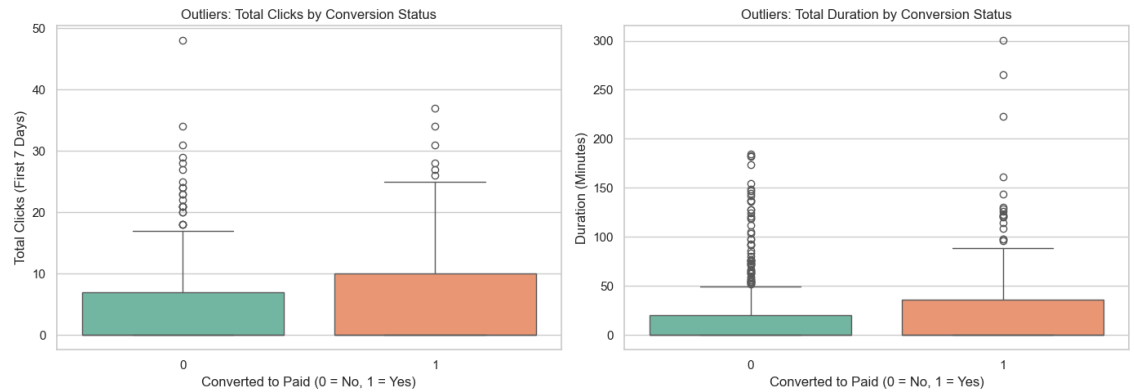
- Since the median for both clicks and duration is exactly 0, it means more than 50% of trial signups have zero log activity in their first week. Most users register and immediately abandoned the product.

b) Distributions



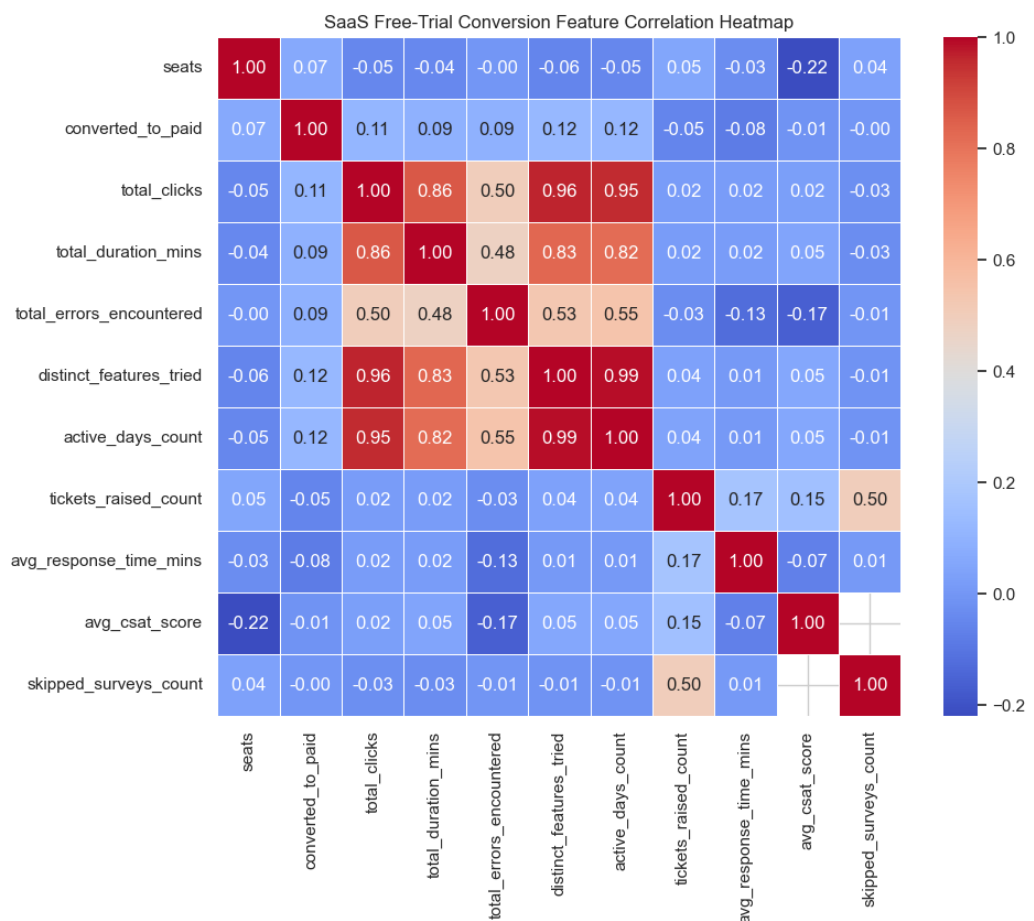
- The vast majority of accounts register zero or near-zero engagement (logging in briefly and abandoning), while a small minority of high-intent "power users" generate long session durations, high click counts, and deep feature exploration.
- Because extreme right-skewness violates the assumption of normality and inflates the standard deviation, relying on the mean can produce misleading conclusions.
- **Medians** and non-parametric tests—specifically the **Mann-Whitney U Test**—were selected to evaluate differences between converters and non-converters.

c) Outliers



- We have non-converters who logged up to 50 clicks and 200 minutes (3.3 hours) of usage in their first week, yet chose not to pay. This directly supports our hypothesis that they get a massive amount of free value during the trial to finish a one-time task and had no reason to subscribe.

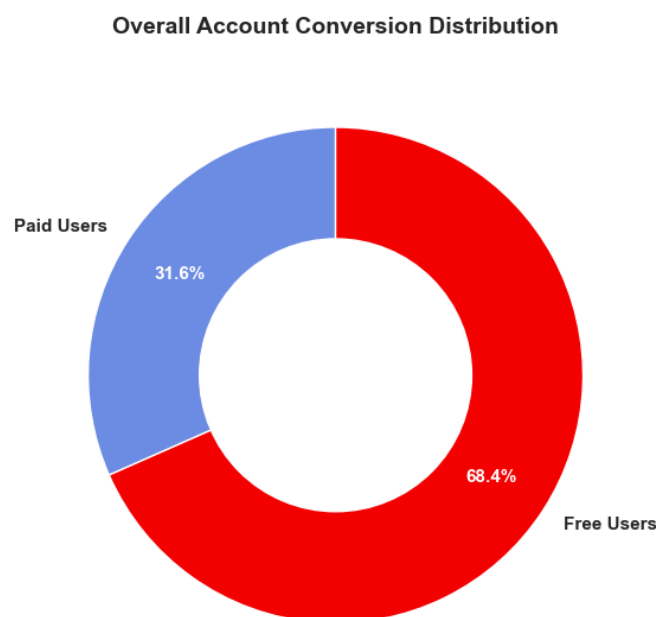
d) Correlation



- **total_clicks** and **total_duration_mins** are almost perfectly correlated. Keeping both would cause redundancy.
- A negative correlation between **converted_to_paid** and **avg_csat_score** means that users who give lower satisfaction scores are actually more likely to convert. This indicates that users who are interested in software ignore minor errors and they still buy because they need it.
- A high correlation between **tickets_raised_counts** and **skipped_surveys_count** means the more tickets a user submits, the more they skip the feedback survey, this directly indicates that consumers feel overwhelmed by excessive or irrelevant brand messages.

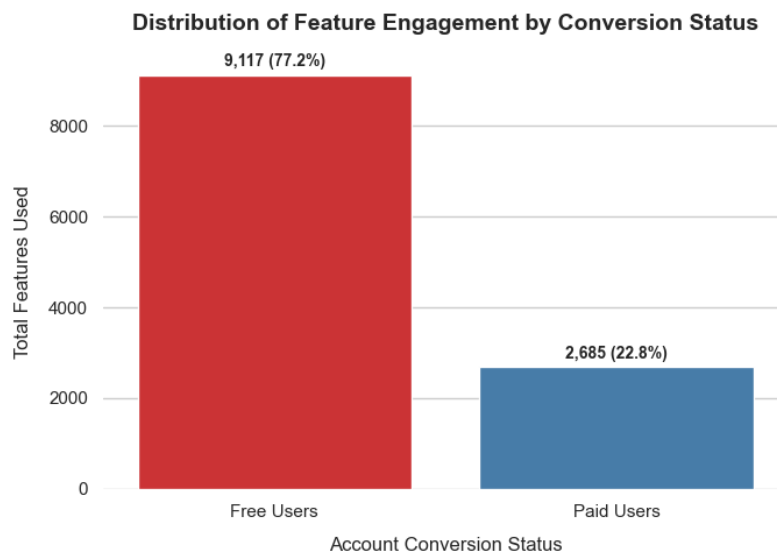
2. Research Questions

- *What is the fundamental trial-to-paid conversion baseline for the platform?*



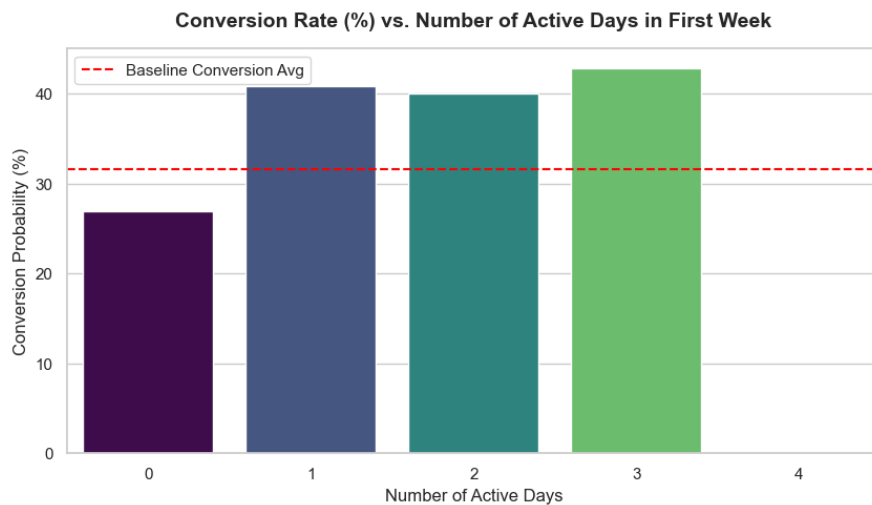
- Out of 500 trial accounts, **158 accounts converted (31.60%)**, while **342 dropped off (68.40%)**. This establishes a baseline churn threshold against which all subsequent behavioral and operational segments were evaluated.

- *Do session duration and overall click volume correlate with trial conversion? Are user activity metrics normally distributed?*
 - Time and click volume directly correlate with trial success. Converted accounts exhibited significantly higher total session durations (mean of **28.0 minutes** vs. **19.27 minutes** for non-converters).
 - Users spending less than **20 minutes** in the app during their trial window represent an extreme drop-off risk.
 - A **Mann-Whitney U Test** ($P = 0.0036$) confirmed that converting users interact with the product significantly more than non-converters, even though the median click counts for both cohorts hover near 0.0 due to a high volume of inactive signups.
- *Does exploring multiple distinct features improve conversion, or do non-converters extract value without upgrading?*



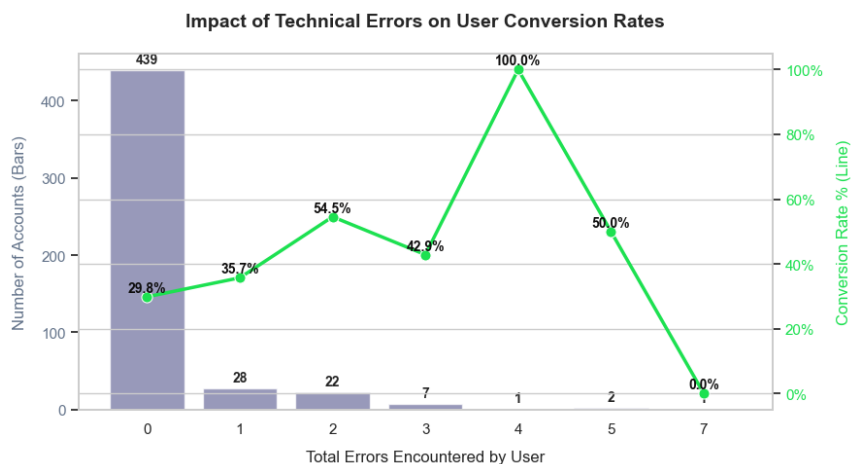
- Exploring **1 to 2 distinct features** increases conversion probability from **26.91% to over 40.63%**.
- Non-converters accounted for **9,117 usage events (77.25% of total platform activity)**. Specifically, feature_20 generated the highest engagement among non-converters (averaging **49.0 clicks** and **82.99 minutes**).

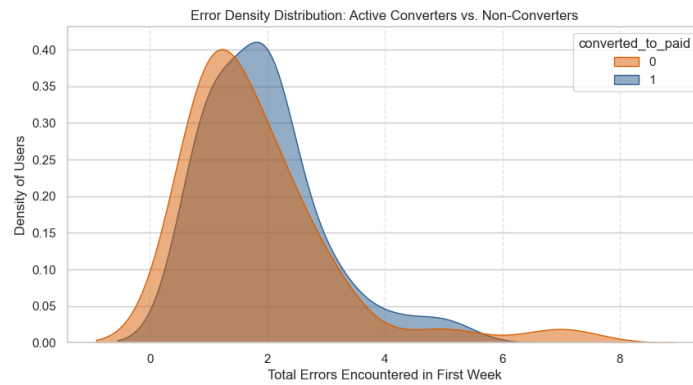
- *How does active logging frequency impact conversion likelihood, and does engagement drop between Week 1 and Week 2?*



- Moving an account from **0 active days (26.91% conversion)** to **1 active day (40.77% conversion)** produces a **+13.86 percentage point lift**. Engagement beyond 2–3 active days hovers around 40%–42%.
- A **Paired T-Test** comparing Week 1 vs. Week 2 clicks for non-converting accounts yielded a **T-Statistic of 0.0581** ($P = 0.9537$).
- We fail to reject the null hypothesis (H_0). Non-converters do not "lose steam" mid-trial; their activity remains flat and low from Day 1.

- *Is technical error density a primary driver of customer drop-off?*



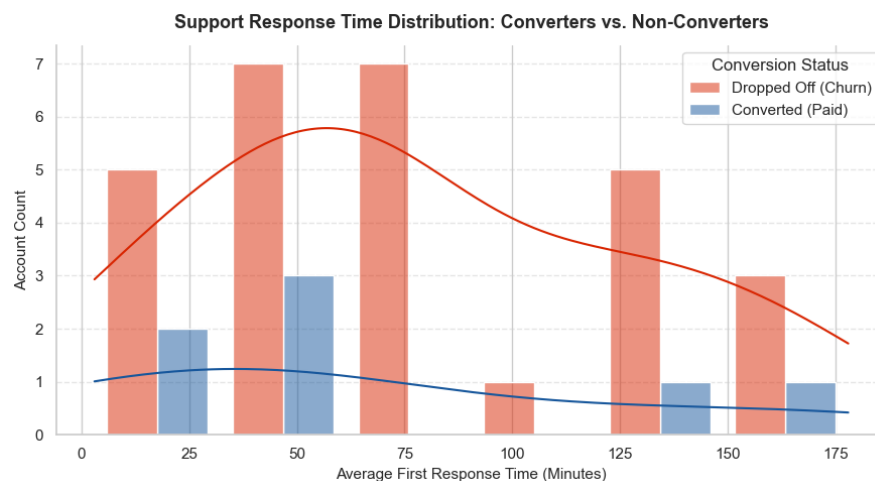


- Accounts encountering **1–2 errors** converted at higher rates (**35.71% to 54.55%**) than zero-error accounts (**29.84%**).
- A **Mann-Whitney U Test** yielded $U = 25025.50, P = 0.0196$.
- Data shows that beta features are not responsible for early trial churn or onboarding errors.
- We reject H_0 , but with an unexpected direction: **converters encounter significantly more errors than non-converters**, because high-intent power users explore the product deeply during their trial, so they ignore minor bugs and see the product's value.

○ *Does submitting a support ticket signal a risk of trial failure?*

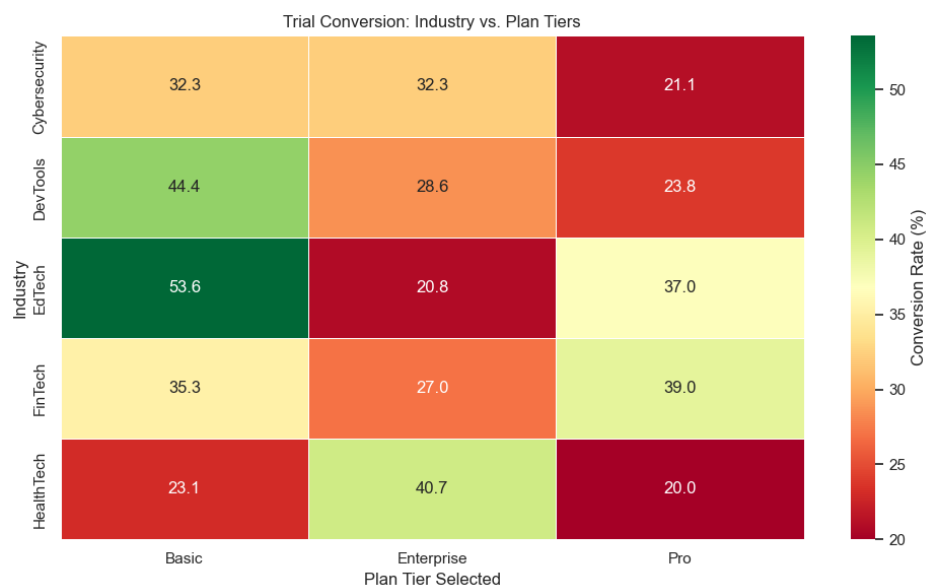
- Out of 34 accounts that raised a support ticket, **28 accounts dropped off (82.35% churn risk)**.
- Opening a support ticket during a short 14-day trial indicates that the user encountered a critical onboarding blocker that broke their momentum.

○ *Do support response delays drive trial drop-offs?*



- A **Mann-Whitney U Test** comparing resolution delays for converters vs. non-converters yielded $U = 115.50, P = 0.2416$.
 - Statistical testing shows that **support response time was not the bottleneck causing users to leave**. Users churned regardless of how fast or slow support replied, indicating that friction was driven by fundamental product issues (like pricing or feature complexity) rather than support responsiveness.
- *Do high CSAT ratings on support tickets predict successful conversions?*
- Accounts rating support 5.0 out of 5.0 had a **0.00% conversion rate** (0 for 2), while 4.0 ratings converted at **22.22%**.
 - CSAT scores do not predict trial conversion. **Helpful customer support cannot overcome underlying product friction**, such as pricing misalignment or missing feature requirements.
- *Does support resolution delay negatively impact CSAT scores?*
- A **Pearson Correlation Test** yielded $r = -0.0717, P = 0.7336$.
 - Pearson Correlation Test shows that, a slight delay in first response time does not drive down CSAT scores, nor does a faster response automatically guarantee higher ratings.
- *Are users who skip feedback surveys more likely to churn?*
- A **Chi-Square Test of Independence** yielded $\chi^2 = 0.2188, P = 0.6400$.
 - We fail to reject H_0 . Survey participation is neutral with respect to conversion; skipping a survey is not an indicator of user dissatisfaction or imminent churn.

- Which referral sources deliver the highest volume and highest-converting leads?
 - **Organic Search:** Highest volume (**114 signups**) and highest conversion rate (**39.47%**).
 - **Paid Ads:** Strong performance with **98 signups** and a **36.73% conversion rate**.
 - **Events & Partners:** Lower conversion efficiency (**26.04%** and **25.84%** respectively).
- How do conversion rates vary across different industry verticals and plan tiers?



- **Basic Plan:** Primary conversion driver across verticals, peaking in **EdTech Basic (53.6%)** and **DevTools Basic (44.4%)**.
- **Enterprise Plan:** Performs strongly in **HealthTech Enterprise (40.7%)**, but drops significantly in **EdTech Enterprise (20.8%)**.
- **Pro Plan:** Consistently underperforms across **Cybersecurity Pro (21.1%)**, **DevTools Pro (23.8%)**, and **HealthTech Pro (20.0%)**.

3. Statistical Hypothesis Testing & Findings

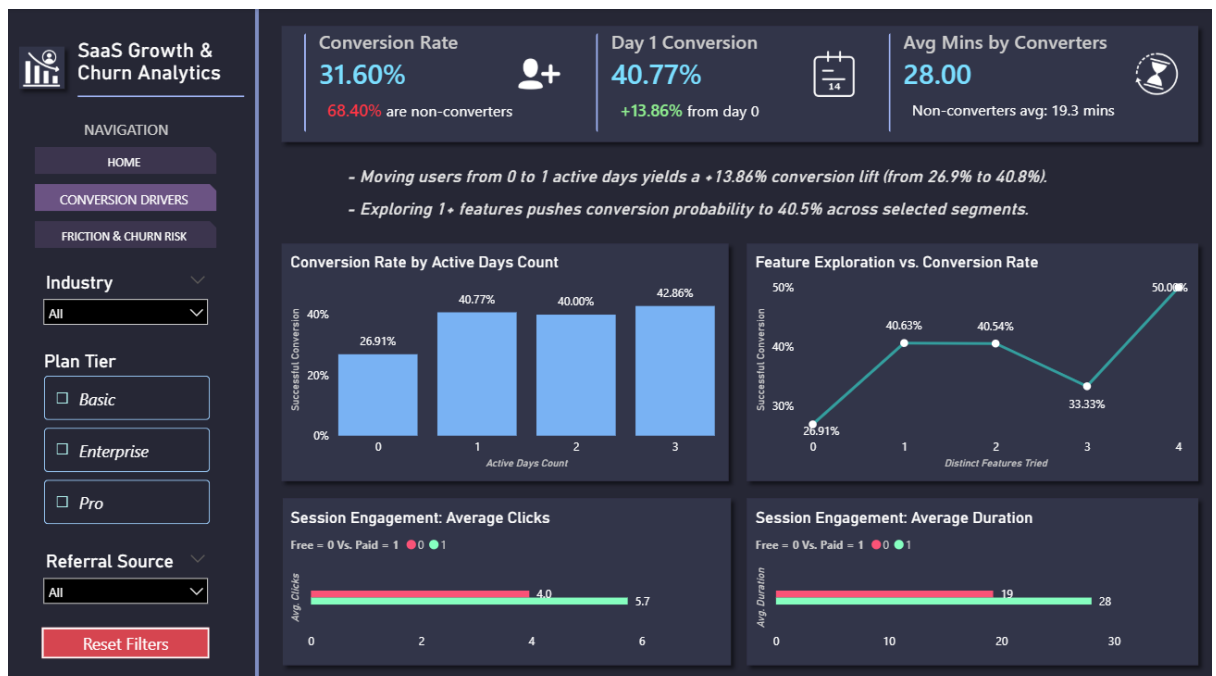
Hypothesis / Experiment	Test Applied	Result (P-Value)	Business Insight
Week 1 vs. Week 2 Usage Drop-off	Paired T-Test	$P = 0.9537$	Failed to reject H_0 . Non-converters stay flat and inactive across both weeks rather than losing steam over time.
High Usage Intensity Drives Conversion	Mann-Whitney U	$P = 0.0036$	Reject H_0 . Converters log significantly higher session duration (28.0 mins vs 19.3 mins).
Technical Errors Drive Churn	Mann-Whitney U	$P = 0.0196$	Error Paradox: Converters hit <i>more</i> errors because high-intent power users explore deeper, whereas churned users log off before hitting errors.
Support Delays Drive Conversion Loss	Mann-Whitney U	$P = 0.2416$	Failed to reject H_0 . Support resolution delays are not significantly higher for non-converters. Ticket creation itself is the friction event.
Support Response Delay vs CSAT	Pearson r	$r = -0.0717$ ($P = 0.7336$)	No correlation between response speed and CSAT score. Resolution quality matters far more than raw response time.
Survey Avoidance Correlates with Churn	Chi-Square (χ^2)	$P = 0.6400$	Failed to reject H_0 . Skipping feedback surveys is neutral and does not indicate churn intent.

Phase 3: Interactive Storytelling (Power BI)

The final phase was building a decision-making tool for non-technical users.

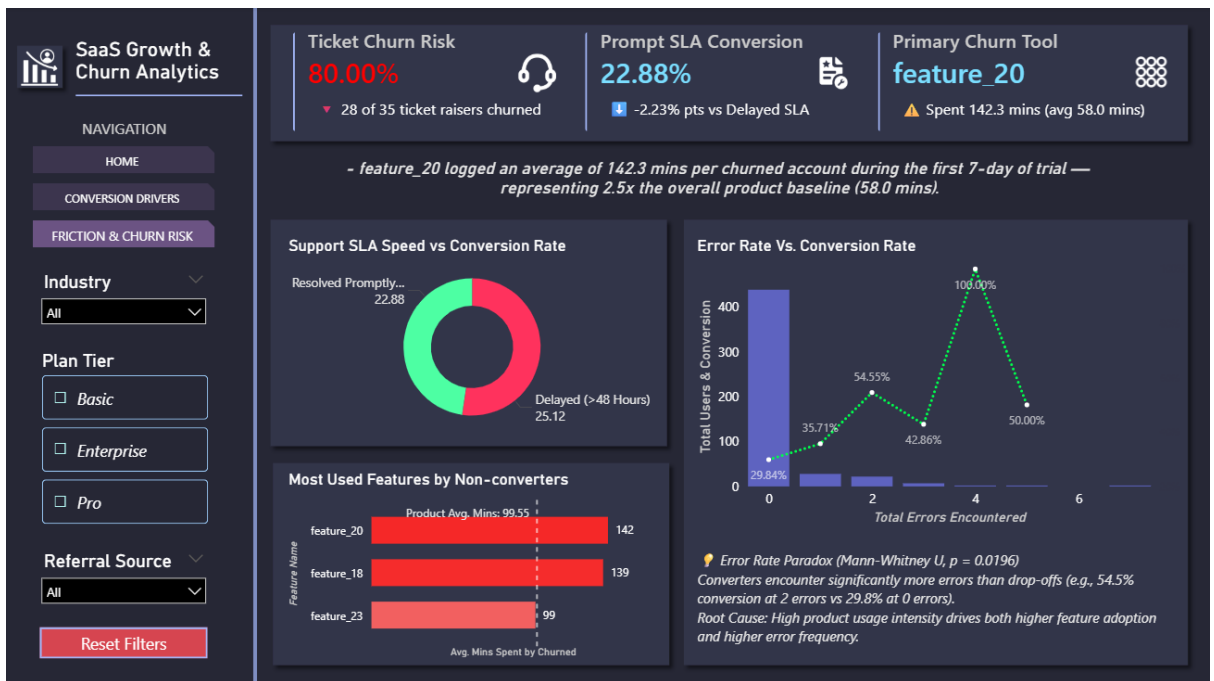
❖ Page 1: Conversion Drivers

- **Slicers:** Slicers for Industry, Plan Tier, and Referral Source.
- **KPI Scorecards (Dynamic Subtitles Included):**
 - **Baseline Conversion Rate:** 31.60%
 - **Day-1 Activated Conversion Rate:** 40.77%
 - **Avg Active Session Duration:** 28.0 mins
- **Chart 1 (Clustered Column):** Active Days (0 to 4) vs. Conversion Rate %. Highlights the +13.86% jump on Day 1.
- **Chart 2 (Line Chart):** Distinct Features Tried (0 to 4) vs. Conversion Rate %. Shows conversion stabilizing above 40% after 1–2 features.
- **Chart 3 (Clustered Bar):** Session engagement for average clicks and average duration.
- **Dynamic Smart Card:** Uses DAX string concatenation to auto-generate contextual insights based on slicer selections.



❖ Page 2: Friction & Churn Risk

- **Top KPI Cards (Dynamic Subtitles Included):**
 - **Ticket Churn Risk:** 80.00%
 - **Prompt SLA Lift:** 22.88%
 - **Top High-Risk Feature:** feature_20
- **Chart 1 (Dount Chart):** Resolution Speed ($\leq 48h$ vs $>48h$) vs. Conversion Rate %.
- **Chart 2 (Clustered Bar):** Avg Mins Spent by Churned Users on feature_20, feature_18, and feature_23.
- **Chart 3 (Line & Clustered Column):** Error Rate vs. Conversion Rate.



Strategic Recommendations

- Trigger automated email and in-app push notifications, contextual tooltips, and banner announcements when an account registers zero active clicks within 24 hours of signing up.
- Build guided walkthroughs that force new users to interact with at least 2 core features during their first login to cross the 20-minute engagement milestone.
- Invest in contextual in-app guidance, embedding instant AI chatbots, and fixing core product bugs so users never hit blockers that force them to submit a ticket.
- Place usage limits on high-utility features like feature_20 and feature_18. Allow 3–5 free uses before requiring a paid upgrade, converting high-usage non-converters into paying subscribers.
- Restructure the Pro plan's features and price point to create a clearer bridge between Basic and Enterprise.
- Shift marketing budget away from events and partnerships (which suffer ~28% churn) toward high-intent Organic and Paid Search channels.